NCH-POUND

MIL-PRF-49471/10(ER)
2 June 1995

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, HIGH PERFORMANCE, BA-X112/U

This specification is approved for use within Army Research Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications specified in the and Standards (DODISS) solicitation: MIL-PRF-49471(ER).

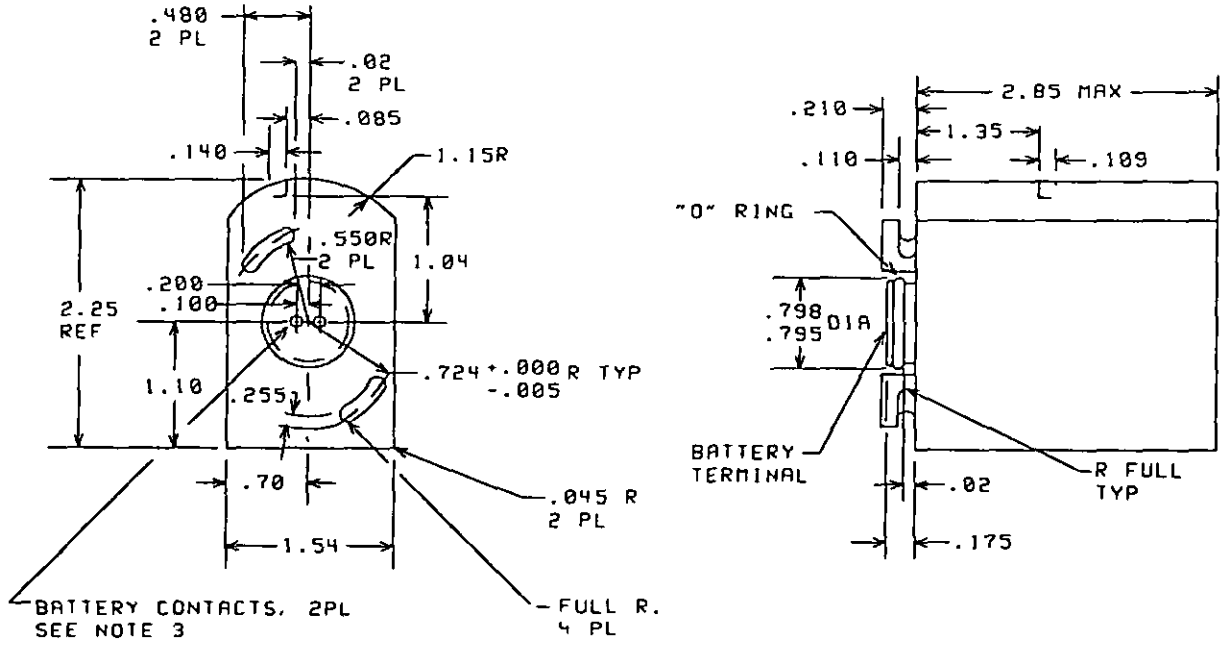


Figure 1. Battery dimensions.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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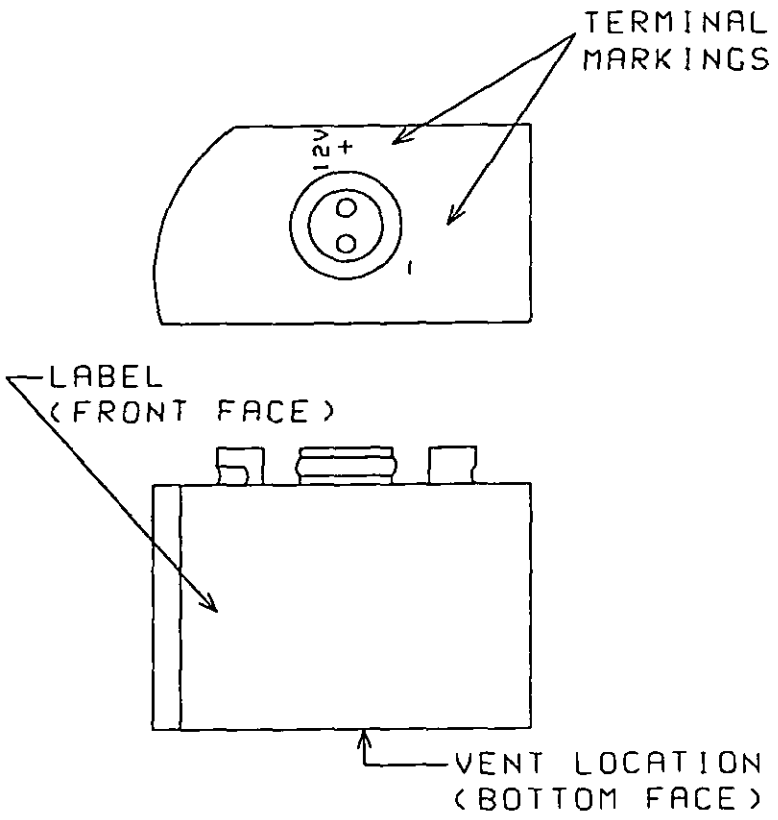


Figure 2. Terminal markings.

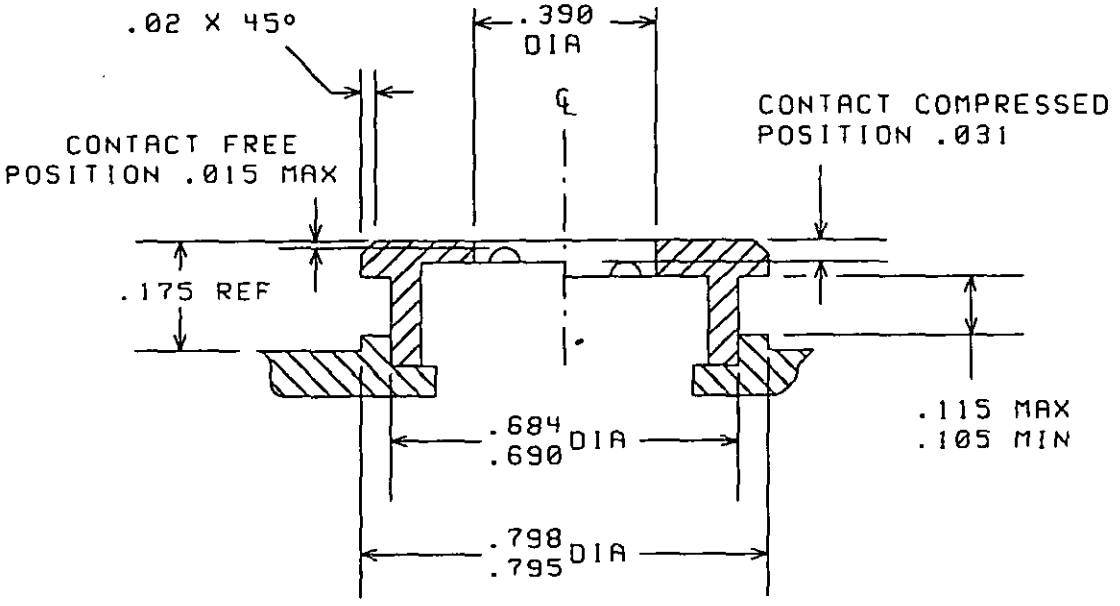


Figure 3. Contact Configuration.

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NOTES:

1. All dimensions are in inches.

2. Tolerances are: 2 place decimals ±.01.
   3 place decimals ±.005.

3. The contacts shall not protrude above the surface of the battery terminal. The contact's diameter shall be .109±.005 inch and the contact surface shall be smooth.

REQUIREMENTS:

Dimensions and configuration: See Figure 1 and Figure 3.

Maximum voltage: 12.20 volts.

Minimum voltage for capacity: 8.00 volts.

Charge protection: Applicable.

Safety features: Applicable.

High temperature device: Applicable.

Complete discharge device: Applicable.

Terminals: 2 pin contact, coupler interface, "O" ring sealed (see Figure 1 and 3).

Battery contacts: The contacts shall be constructed from tubular, oval head rivets conforming to MS16535-11. The contacts shall be nickel plated, 0.0010 inch minimum thickness. Then the contacts shall be gold plated, 0.00010 inch minimum thickness, approximately 0.0456 square inches. The contacts shall be mounted on a spring-loaded surface. The contacts shall require a minimum mating force of 15 ounces per contact.

Weight (maximum): 0.40 pounds (180 gms).

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Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in Hours to 8.00 Volts
I	3.00
L	1.50
H	5.50
HT	5.30
LT	1.30
IT	2.80

Initial voltage delay: During the I, H, HT, and IT tests, initial voltages below the 8.00 volt minimum voltage cannot exceed a 5.0 second duration. During L and LT discharge tests initial voltage below 8.00 volts cannot exceed a 60.0 second duration.

Venting mechanism: The battery case shall incorporate a venting mechanism that shall prevent the battery case from exploding or rupturing into fragments when cell venting occurs. The venting mechanism shall be located at the bottom face (see Figure 2.0). Venting mechanism shall not become separated from the battery case when activated by cell venting.

O-ring seal: The battery when subjected to the O-ring seal test shall not increase in weight by more than 0.1 grams and the O-ring shall prevent the water from contacting the battery contacts and the battery terminal face. There shall be no sign of corrosion on the battery contacts.

METHOD OF EXAMINATION AND TEST:

Capacity tests:

(1) Storage:

Details on storage conditions for all specified capacity tests are described in the basic specification.

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(2) Discharge:

Cells (forced discharge): One cell shall be discharged at 0.65 amperes at 70±5°F to 2/3 of open-circuit voltage. The discharged cell shall then be connected in series to three fresh cells. Each string shall be discharged at 0.65 amperes at 70±5°F to 2/3 open-circuit voltage of the three fresh cells.

Batteries: For all capacity tests, except the H and HT, the batteries shall be discharged at 0.65 ampere constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through 0.65 ampere constant current for five minutes followed by open-circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches zero volts.

Closed-circuit voltage: Load resistance of 20 ohms shall be used between battery terminals. Voltage shall be above 8.0 volts within 5 seconds.

Venting mechanism: The battery case venting mechanism shall be monitored during the safety feature test. During cell venting the battery venting mechanism shall release the pressure build up in the battery case and shall meet the performance requirements specified herein.

O-ring seal test: The samples used for the I and L tests of the first article inspection and group D inspection shall be used. These samples shall be selected prior to the capacity tests. The weight of each battery sample shall be measured and recorded. The samples shall be mated to a Motorola part number 64-P21451J001 coupler plate or equal. The coupler plate shall be waterproofed, except at the battery/ coupler plate interface, to prevent water from contacting the battery connector contacts. The battery/plate assemblies shall be immersed in a synthetic solution containing 3.6 percent by weight sea salt in accordance with ASTM-D-1141. The solution with the battery samples shall be pressurized to simulate two foot immersion for 24 hours and the pressure shall be raised to simulate 50 foot immersion for 8 hours. The battery samples shall be removed from the solution and separated from the coupler plate. The battery contacts shall be inspected for evidence of water. The batteries shall be towel dried and left to air dry for 15 minutes. The batteries shall be weighed and shall meet the O-ring seal requirements specified.

Custodian:
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Preparing Activity:
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(Project 6135-A274-10)

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